From Raw Nuclei Images to an Auditable Report

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GitHub Repository: https://github.com/mdsajidhasan510/Assignment3-AI-Imaging

What I built and why

My allocated modality was the fluorescence microscopy of stained nuclei, with the task being to push images through a pipeline: clean the data, have a local image model explain an image, extract classical features and have a text model interpret pure numbers and then train a little U-Net (Ronneberger, Fischer and Brox, 2015), and then glue them all together to write a JSON record and a brief paragraph for each unfamiliar test image. All is executed locally on a T4 Colab using Ollama. The rule I followed with all these was very simple: a language model can never say anything that I can't check, and so my measured values go with each record that it can say.

The data, and what the metadata says before any pixels

The set contains 80 training images, 20 validation images, 12 held-out test images, each with an accurate binary mask and a metadata table containing the correct object counts and a density regime per image. First I read the metadata: training images are 40 normal, 14 sparse, 13 dense and 13 clustered, with numbers of nuclei ranging from 5 to 85. All images were first converted to grayscale and all images were first reduced to a common size 256 x 256 grid. As seen in Figure 1, three training images with their respective masks as contours were used; Figure 2 is the pooled intensity histogram of all three images; 93.17% of the pixels are below 0.2 intensity.

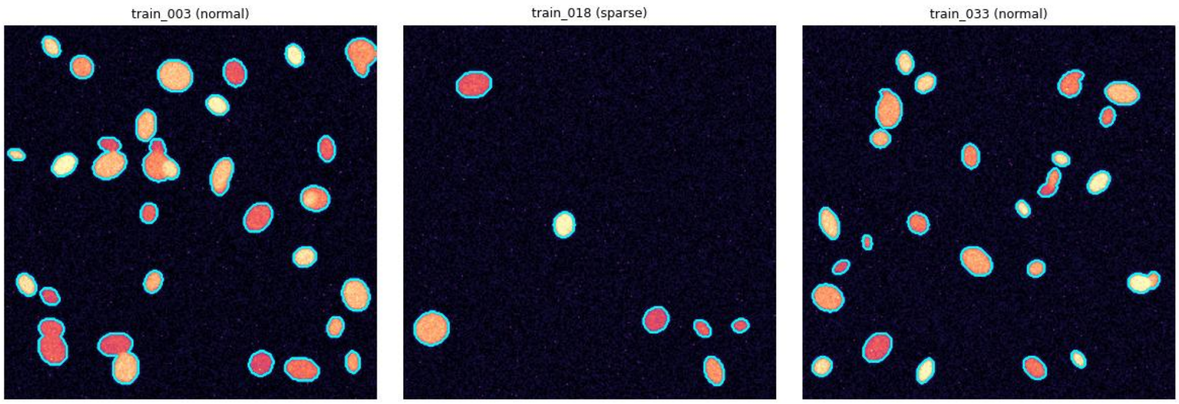

Figure 1. Training images with ground-truth mask contours and their density regimes.

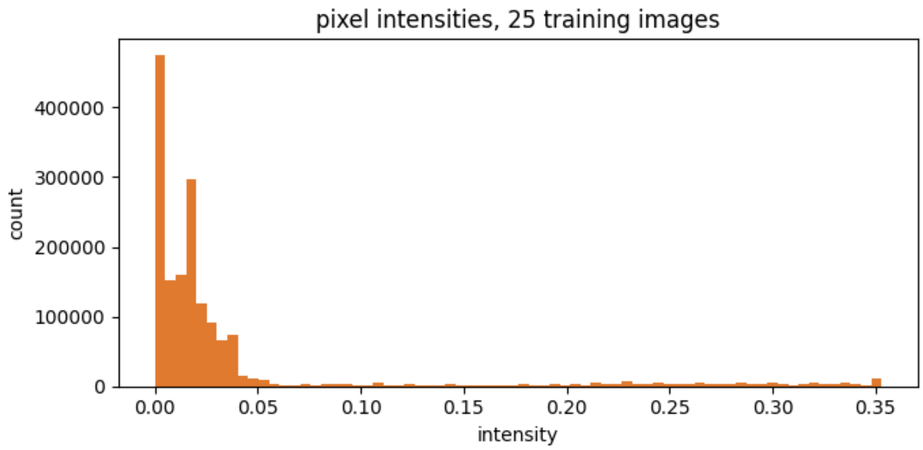

Figure 2. Pooled pixel intensities over 25 training images.

Nearly everything is dark background and the nuclei occupy a narrow bright shoulder at the top of the range, which is close to ideal for a global threshold. That sets a high bar for the U-Net: Otsu (1979) costs nothing and was always going to score well here.

Getting a vision model to behave

I tried to use llama3.2-vision, but it is almost what you'd expect, it is doing exactly what the module team warned as it is using unknown model architecture: 'mllama' and thus switching to qwen2.5vl:7b (Bai et al., 2025). The lazy prompt 'What is in this picture?', given on train_012, returned as a chatty paragraph confidently explaining dark-field and phase-contrast optics of which it can know nothing. I have it tuned using the prompt below: describe, don't diagnose, fixed JSON keys, explicit way out if unsure.

```
Look at this microscopy image and describe it factually.
Do not diagnose anything and do not guess at clinical meaning. Stick to what is visible.
Answer with one JSON object only, using these keys:
"modality" (what kind of imaging this looks like),
"tissue type" (or "uncertain" if you cannot tell),
"notable features" (a short list of things you can actually see),
"image quality" ("good", "acceptable" or "poor").
If unsure about a key, put "uncertain". No text outside the JSON.
```

At temperature 0 it returned the clean JSON: modality 'microscopy', tissue_type 'uncertain', four concrete features and quality 'acceptable'. I then repeated the same prompt at three different temperatures (0.2, 0.7 and 1.0) and no two runs were the same even with 0.2 – at that temperature, the list of features was different between runs. Any nonzero temperature returns the same value upon repeated sampling and the output of the prompt is the temperature 0 one, the prompt itself being the artefact of interest.

Classical features, and a model that only sees numbers

Otsu plus morphological cleanup segmented train_012 into 10 objects (Figure 3), with areas 183 to 544 px (mean 305), mean eccentricity 0.63 and mean solidity 0.94 from the regionprops table. Only a one-line numeric summary went to the text model llama3.2, using this prompt:

```
Below is a set of measurements from an automated analysis of one microscopy
image. You have not seen the image. Work only from the numbers.
Give me:
DESCRIPTION: a plain paragraph, three sentences or fewer, describing the scene.
RECORD: one JSON object with keys "n_objects" (int), "density_class" (one of
"sparse", "moderate", "dense"), "shape_regularity" ("regular" or "irregular"),
"quality_flag" ("ok" or "review").
Use exactly those two labels, DESCRIPTION: then RECORD:. Never add numbers I did not give you.
```

It gave a sensible three sentence description and a record: n_objects 10, density_class 'moderate', shape_regularity 'irregular', quality_flag 'ok'. The three-way check is: Otsu saw 10, LLM saw 10, metadata truth is 12. The model was able to copy my number exactly and the model lost the two nuclei, which was done earlier, when the global threshold was used to merge touching pairs. The split does make a difference: it was the only stage there was no error on.

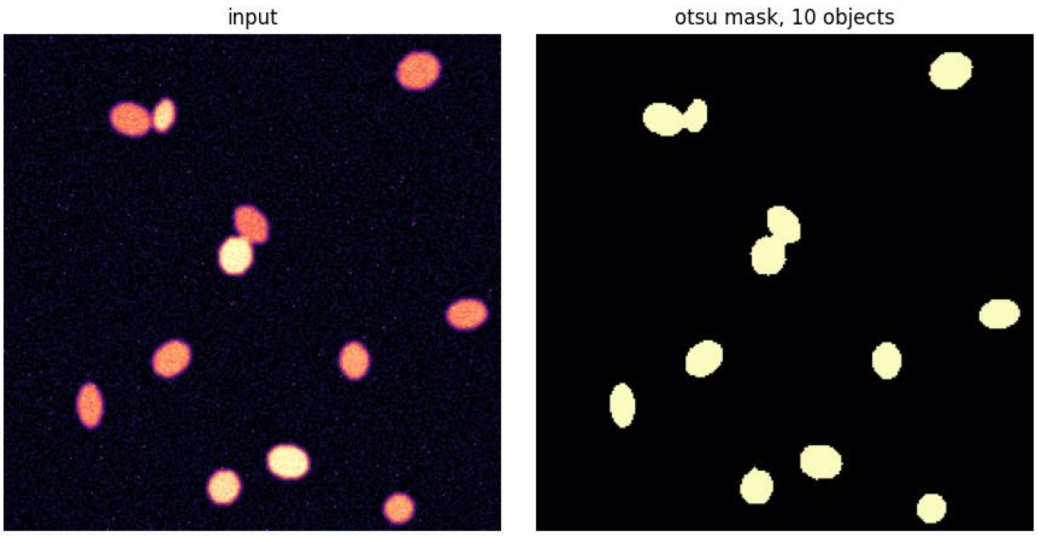

Figure 3. Input (left) and Otsu mask (right) for train_012, 10 objects after cleanup.

The U-Net

The network was trained using a 3-level, width 32 (1.93M parameters) U-Net and a plain soft Dice loss, as it is preferable when the classes are heavily imbalanced (93% background), and Dice optimises for overlap directly without any weighting fiddle (Milletari, Navab and Ahmadi, 2016). Batch size: 4, Adam: 5e-4, flips + right-angle rotations, 12 epochs. The curves are shown in Figure 4: validation Dice is at 0.4247, exceeds 0.94 by the second epoch and reaches 0.9938 with mean IoU of 0.9877 on 20 validation images. Three of them are shown in figure 5 with their input, truth and prediction.

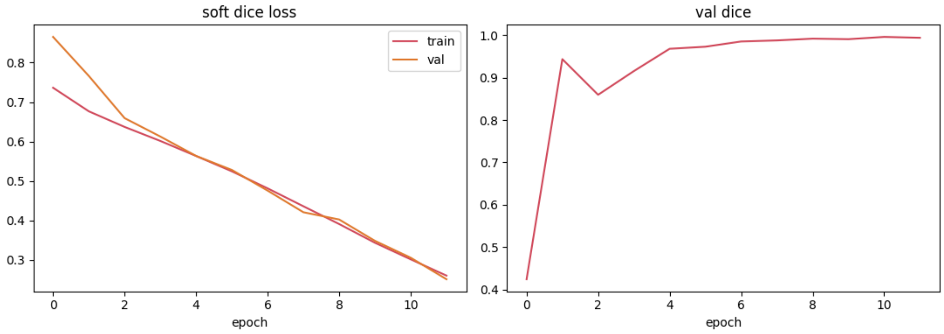

Figure 4. Soft Dice loss and validation Dice over 12 epochs.

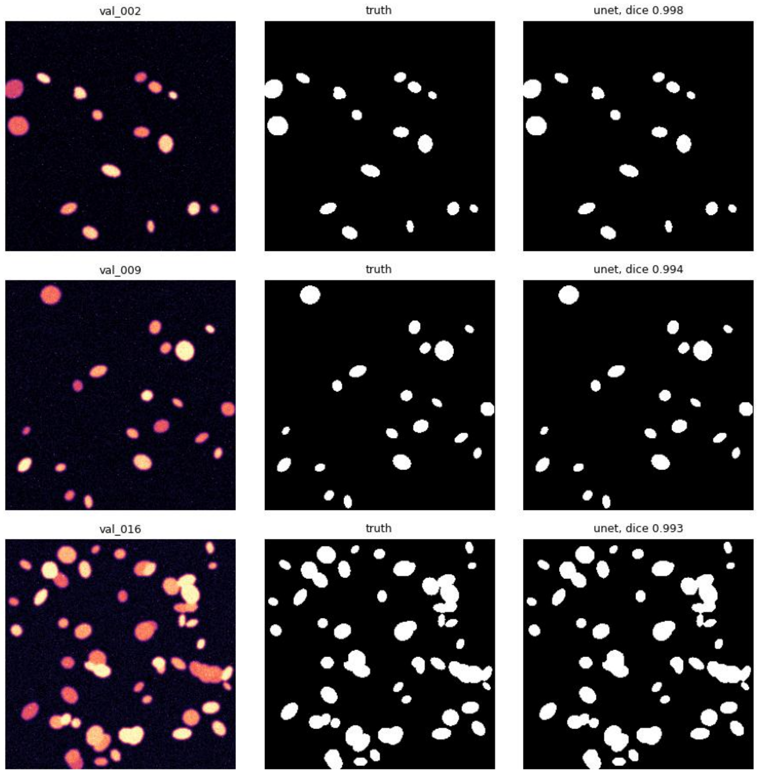

Figure 5. Validation panels with per-image Dice.

Head to head with Otsu on all 20 validation images, the U-Net wins 20 of 20: means 0.9938 against 0.9775. The best win is val_002 (+0.0213 Dice) and the narrowest is val_012 (+0.0109), so even where Otsu gets closest the network stays ahead by about a hundredth of Dice.

The full pipeline on the twelve unseen images

Each test image runs through U-Net mask, cleanup, regionprops, numeric summary, then one temperature 0 call with the prompt below. Empty masks skip the model entirely, and my measured count and mean area ride along in every record as audit columns.

```
You write structured records for an automated microscopy report.
Input: measurements from one image, given below. That is all you know.
Output, in this exact shape:
RECORD:
{"image_id": "...", "n_objects": <int>, "mean_area": <number>, "density_class":
"sparse"|"moderate"|"dense", "quality_flag": "ok"|"review"}
REPORT:
A short paragraph, two to four sentences, plain language, no diagnosis, no numbers
that are not in the measurements. Flag quality_flag as "review" if anything looks odd.
```

The aggregated CSV (test_set_records.csv) has 12 rows. For example, test_003: {"image_id": "test_003", "n_objects": 19, "mean_area": 207, "density_class": "dense", "quality_flag": "ok", "n_objects_measured": 19, "mean_area_measured": 206.7} and the report paragraph would be: "The image contains a dense cluster of objects. The objects appear to be well-defined and evenly sized, with a moderate level of intensity. Overall, the image quality is good. I had 12 records of counts that matched my measurements. The categorical fields are another matter — the model was given the label "dense" for each image, even test_000, with 8 objects. There are numbers that it was given, and there are judgements that it was asked to make, that survive.

Extension: which loss trains the best U-Net here?

Same data, same width-16 architecture, same seed and a short budget of just three epochs, and three losses. The results (Figure 6, Table 1) surprised me: plain BCE won with 0.9005 validation Dice, followed by BCE plus Dice with 0.8874, and Dice alone last with 0.8396. However, the Dice loss for 12 epoches gave 0.9938 on my main model. The honest read from the paper is that the loss ranking is a function of budget; BCE will get closest to convergence from scratch, but Dice will catch up and surpass in epochs with more budget. If I had an endpoint ablation done after 6 epochs it would have taken me down the wrong path for the entire run.

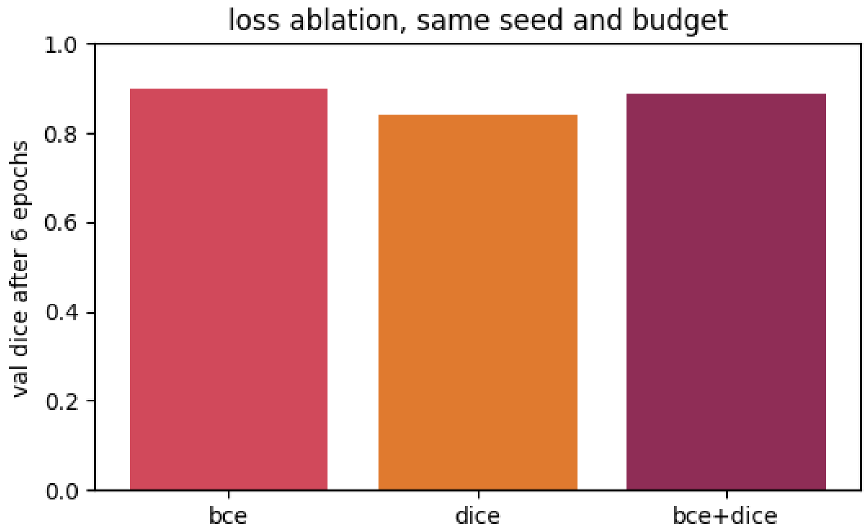

Figure 6. Validation Dice after 6 epochs under each loss, same seed and budget.

Setup	Loss	Epochs	Val Dice	Val IoU
Otsu baseline	none	n/a	0.9775	n/a
Main U-Net (w=32)	Dice	12	0.9938	0.9877
Ablation U-Net (w=16)	BCE	6	0.9005	n/a
Ablation U-Net (w=16)	Dice	6	0.8396	n/a
Ablation U-Net (w=16)	BCE + Dice	6	0.8874	n/a

Table 1. Every segmentation result in this study, on the 20 validation images.

The five questions, answered straight

Useful against trustworthy: Q1. The direct description is easier to read; it flows and provides context a table won't. Description of each number is more reliable as it references a regionprops row and I checked its number against measurements and ground truth. The lazy vision had answered the invented optical physics, the numbers-first vision could not, since it never saw the image. For pipeline, it's no slam dunk.

This is Q2, U-Net vs Otsu. Yes, the U-Net improved: 0.9938 mean Dice vs the 0.9775 mean Dice of the previous U-Net and all 20 validation images were won. The best example of its success is val_002 (+0.0213), a dense image where nuclei touching is rewarded over a global threshold. Otsu never beat the network outright, the best image is val_012, a sparse clean frame where the thresholding already works pretty much perfectly. It is a synthetic data Otsu where it is good, with the edge of the network being the boundary precision and not detection.

Q3 : what do the scores indicate? Dice 0.9938 and IoU 0.9877 indicate that there was only a small difference between the predicted and the true masks, with IoU being stricter so it will be lower. What is still left is some 1 or 2 pixel halos around the nucleus boundaries and the worst validation images are clustered images (images with the highest proportion of boundary pixels in the mask). Early-epoch Dice also swings hard (0.42 to 0.94 in one epoch) because the hard threshold is applied to a significant number of pixels at a time when the logits are still close to the boundary.

Hallucination and JSON in Q4. The vision step can hallucinate modality and physics, and did with the lazy prompt. The text steps can hallucinate categories and did: 'dense' for all test images, no matter how many they were. What reduced the risk: forcing description over diagnosis, an explicit ‘uncertain’ option, temperature 0, models that never see pixels at the interpretation stage (Ji et al., 2023) and audit columns beside every model field. The JSON is the source of truth because it can be mechanically diffed with measurements, and that is how I know that counts were perfect in 12 of 12, but density was close to wrong everywhere.

Would I use it clinically? (Q5) No. These are synthetic, clean and single style; real nuclei imaging is a lot messier (Caicedo et al., 2019) and even my LLM stage labels density wrong on easy inputs. The one thing that would help the most in making this more trustworthy is to evaluate on real annotated microscopy first, before evaluating on any other thing; otherwise, all the numbers above are upper bounds, not estimates. Then I would remove all of the categorical decisions from the LLM and calculate it using the measured features.

References

Bai, S., Chen, K., Liu, X. et al. (2025) 'Qwen2.5-VL technical report', arXiv preprint arXiv:2502.13923. Available at: https://arxiv.org/abs/2502.13923 (Accessed: 17 August 2026).

Caicedo, J.C., Goodman, A., Karhohs, K.W. et al. (2019) 'Nucleus segmentation across imaging experiments: the 2018 Data Science Bowl', Nature Methods, 16(12), pp. 1247-1253. Available at: https://doi.org/10.1038/s41592-019-0612-7 (Accessed: 17 August 2026).

Ji, Z., Lee, N., Frieske, R. et al. (2023) 'Survey of hallucination in natural language generation', ACM Computing Surveys, 55(12), pp. 1-38. Available at: https://doi.org/10.1145/3571730 (Accessed: 17 August 2026).

Milletari, F., Navab, N. and Ahmadi, S.-A. (2016) 'V-Net: fully convolutional neural networks for volumetric medical image segmentation', in 2016 Fourth International Conference on 3D Vision (3DV). IEEE, pp. 565-571. Available at: https://doi.org/10.1109/3DV.2016.79 (Accessed: 17 August 2026).

Otsu, N. (1979) 'A threshold selection method from gray-level histograms', IEEE Transactions on Systems, Man, and Cybernetics, 9(1), pp. 62-66. Available at: https://doi.org/10.1109/TSMC.1979.4310076 (Accessed: 17 August 2026).

Ronneberger, O., Fischer, P. and Brox, T. (2015) 'U-Net: convolutional networks for biomedical image segmentation', in Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015), LNCS 9351. Cham: Springer, pp. 234-241. Available at: https://doi.org/10.1007/978-3-319-24574-4_28 (Accessed: 17 August 2026).